

Project Master Knowledge Document: PAKU

1. Project Overview & Identity

- **Project Name:** PERSONALIZED AUTONOMOUS KNOWLEDGE UNIT (PAKU).
- **Project Type:** 4th Year B.Tech Computer Science and Engineering Major Project.
- **Institution:** The NorthCap University, Gurugram.
- **Core Vision:** To engineer a Self-Adaptive Agentic Personal AI Clone that serves as a proactive digital extension of the user. PAKU autonomously mimics the user's voice, facial expressions, communication style, and decision-making patterns to attend meetings and fulfill academic/professional duties on their behalf.
- **Primary Target Demographic:** Educators, teachers, and academic professionals (with scalability to corporate employees and freelancers).

2. Team & Leadership

- **Project Supervisors:** Dr. Anuradha Dhull (Associate Professor) and Dr. Srishti Sharma (Associate Professor).
- **Core Development Team:**
 - **Nischal Sharma:** Lead for Computer Vision, Edge Computing, Video Cloning Models (face + lip-sync), and Distributed Inference architecture.
 - **Chahat Gupta:** Lead for LLMs, Knowledge Retrieval (RAG), and NLP workflows.
 - **Yashika:** Lead for Speech Processing, Audio Cloning, and Deep Learning.
 - **Nikhil Gupta:** Lead for NLP, OCR integration, and AI Application logic.

3. The Core Problem & Gap Analysis

The transition to digital and remote academic workflows has created immense pressure on educators, forcing them to manage overlapping duties (lectures, administrative meetings, assignment evaluation) simultaneously. Current AI solutions fail to solve this because:

- **Passive Nature:** Existing tools (like Gemini Live, Copilot, Fireflies, Synthesia) are limited to transcription, generic summarization, or scripted playback. They cannot actively participate, make decisions, or represent the user.
- **Severe Privacy Risks:** Commercial AI relies heavily on cloud servers. Uploading sensitive academic material, student data, and confidential institutional discussions to third-party data centers violates privacy principles.
- **Lack of Personalization:** Existing systems lack long-term memory and fail to adapt to a user's evolving communication style or teaching philosophy.

4. System Architecture & Technological Stack

PAKU differentiates itself through a strict **privacy-first, 100% on-device (edge computing) architecture**.

The 5-Layer Workflow

1. **Input Processing Layer:** Captures user/meeting input (voice or text) and identifies intent.

2. **Retrieval & Decision-Making Layer:** Utilizes a Retrieval-Augmented Generation (RAG) framework paired with local vector databases (like ChromaDB or FAISS) to fetch relevant past interactions and context. The reasoning engine uses highly optimized Small Language Models (SLMs), specifically LLaMA 3.2.
3. **Personality Alignment:** The SLM utilizes Low-Rank Adaptation (LoRA) for parameter-efficient fine-tuning, matching the generated response to the user's specific tone, vocabulary, and behavioral patterns.
4. **Multimodal Output Generation:** Converts the text into the user's cloned voice using RVC (Retrieval-based Voice Conversion) and generates synchronized facial animation using diffusion-based models and lip-syncing.
5. **Distributed Inference Layer:** Executes the tasks across available local devices to minimize latency.

Additional Tools & Frameworks

- **Languages:** Python (core pipeline), JavaScript/Node.js (device-level interaction).
- **Machine Learning:** PyTorch, TensorFlow/Keras.
- **Optimization:** GGUF and quantization toolkits for model compression.

5. Core Modules & Current Status

- **Autonomous Meeting Clone (Prototype Phase):** The central agent capable of attending video calls. It integrates voice synthesis, face generation, and contextual reasoning. Currently undergoing latency and quality optimization.
- **Smart Assignment Checker (Completed):** Designed to review large volumes of student submissions. It uses OCR to digitize handwritten assignments, processes the text locally via Ollama, and awards marks based on the presence of key concepts rather than strict spelling, bypassing OCR errors caused by poor handwriting.
- **Lab Manual Evaluator (Completed):** Automates the checking of practical files. The educator uploads an empty template and a "perfect" reference file. The system calculates the difference to understand the required fields and evaluates student submissions against this localized standard.

6. Advanced Innovations & Future Roadmap

- **Heterogeneous Distributed Inference:** PAKU features a hardware-assisted mechanism that partitions workloads across multiple devices (e.g., a mobile phone and a laptop) connected to the same local network. The mobile device handles lightweight tasks (mic/camera access, UI), while resource-heavy inference (LLM, face animation) is offloaded to the laptop via secure local connection/Bluetooth.
- **Predictive Question-Answer Tree (In Development):** Prior to a meeting, the system will scrape the agenda and participant history to predict possible questions. It will pre-render and cache the audio and lip-sync videos for these expected answers. During the meeting, cached responses are deployed instantly, drastically reducing computational load, reserving real-time generation strictly for unpredictable queries.

7. Accolades & Market Impact

- **Hackathon Success:** The PAKU architecture won the Kalasakhi Hackathon, securing ₹88,000 in Google Cloud credits for further research and development.

- **Academic Contribution:** The team has authored a research paper detailing their novel approach to heterogeneous distributed inference on edge devices.
- **Sustainable Development Goals (SDGs):** By eliminating dependency on massive cloud data centers, PAKU's edge-based computing promotes sustainable, energy-efficient digital practices (SDG 11, 12) while enhancing Quality Education (SDG 4) and Decent Work (SDG 8).